

Classification of Binary Ethiopian Scales (Kiñit): Classical Machine-Learning Models and Explainability on the EMIR Dataset

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ABSTRACT

Ethiopian traditional music is organised around pentatonic scales known as Kiñit. This work addresses the binary classification of two Kiñits, Tizita and Bati, on the EMIR dataset using classical, interpretable models. Audio is standardised and described with MFCC (with cepstral mean normalisation) and chroma features, summarised per clip into a 64-dimensional vector. Unsupervised k-means clustering does not separate the scales, indicating that scale is overshadowed by genre and recording variation. Supervised models, however, recover the distinction: an SVM reaches about 80% accuracy and a Random Forest about 86%, well above the 52.9% majority-class baseline. Feature-importance, SHAP and LIME analyses show that classification relies on both timbral and pitch-class features, giving an interpretable account of model behaviour not reported in prior deep-learning work on this dataset.

Index Terms - MIR, Ethiopian music, Kiñit, audio classification, MFCC, chroma, explainable AI, SVM, random forest, SHAP, LIME.

I. INTRODUCTION

Music classification is a core problem in music information retrieval (MIR), but most methods target Western music, leaving many regional traditions under-studied. Ethiopian music is organised around a modal system of pentatonic scales called *Kiñit*. Four are regarded as principal Tizita, Bati, Ambassel and Anchihoie, each defined by a characteristic set of intervals and also associated with distinct stylistic traits, so that Kiñit classification combines scale identification with elements of genre recognition [1].

Automatic Kiñit recognition has practical value for cataloguing and retrieval in digital archives of Ethiopian music, for recommendation, and for musicological study of a tradition with limited computational treatment. This project addresses the binary classification of Tizita versus Bati, reported as among the most distinguishable scales [1], as a tractable first step toward the full four-scale problem.

Early audio classification relied on classical models such as support vector machines and k-nearest neighbours over hand-crafted features, while recent work is dominated by deep networks [2]. On EMIR specifically,

Retta *et al.* [1] benchmarked deep CNNs, and subsequent work on Ethiopian music has used CNN, RNN and LSTM models, reaching high accuracy with MFCC-based features [3]. These approaches are accurate but opaque: they report performance without explaining which acoustic properties drive a decision.

This project takes a complementary direction. It (i) evaluates classical, interpretable classifiers an SVM and a Random Forest suited to the dataset's modest size, and (ii) applies explainability methods (feature importance, SHAP, LIME) to identify the features driving classification. The motivation is that a transparent, lightweight model that performs respectably is valuable both practically and for understanding the scale/genre relationship; the explainability analysis is the main novel element relative to prior EMIR work.

II. METHOD

The pipeline comprises five stages: pre-processing, feature extraction, unsupervised clustering, supervised classification, and explainability.

A. Dataset and Pre-processing

The Ethiopian Music Information Retrieval (EMIR) dataset contains 600 thirty-second recordings spanning Orthodox Tewahedo chants, Azmari songs, and modern secular music, labelled into four Kiñits by five expert judges with high inter-rater agreement (Fleiss kappa = 0.85). [1] This study uses the 306 Tizita (162) and Bati (144) recordings.

Direct inspection revealed that, although documented as 16 kHz, the files are in fact a mixture of 16 kHz and 44.1 kHz. Each recording was therefore passed through the following pre-processing pipeline to produce clean, uniform signals:

1. **Resampling to 16 kHz mono.** All clips were standardized to a common 16 kHz sampling rate and a single channel. Down-sampling was preferred to up-sampling because up-sampling interpolates detail that was never recorded, whereas down-sampling only discards genuine samples; by the Nyquist theorem, 16 kHz preserves spectral content up to 8 kHz, which covers the pitch range relevant to these scales.
2. **DC-offset removal.** The mean of each signal was subtracted to re-centre the waveform on zero,

removing any constant offset, which is a recording artifact with no musical meaning.

3. **Pre-emphasis** ($\alpha = 0.97$). A first-order high-pass filter was applied to boost the naturally weaker high-frequency components, balancing the spectrum so that high-frequency information is fairly represented in the subsequent features.
4. **Peak normalisation**. Each signal was scaled so that its maximum absolute amplitude equals one, removing loudness differences between recordings so that volume cannot act as a spurious cue.

At the next feature-extraction stage, a short-time Fourier transform is computed using a Hann window with a frame size of 2048 samples and a hop length of 512 (75% overlap); the Hann window tapers each frame to suppress spectral leakage.

B. Feature Extraction

Raw EMIR recordings contain both meaningful acoustic information and recording-specific artifacts. Feature extraction aims to preserve characteristics relevant to scale identity while reducing the influence of recording conditions. This helps the classifier learn musical patterns rather than recording-specific characteristics.

Following the feature types established for EMIR [1] and widely used in Ethiopian-music studies [3], two complementary representations are extracted. MFCCs (20 coefficients) capture the spectral shape (timbre) of the signal, while cepstral mean normalization (CMN) reduces the influence of channel characteristics. Chroma features (12 pitch classes) capture pitch-class content directly—the most musically relevant representation of a scale—and, by collapsing octaves, are relatively robust to bandwidth differences.

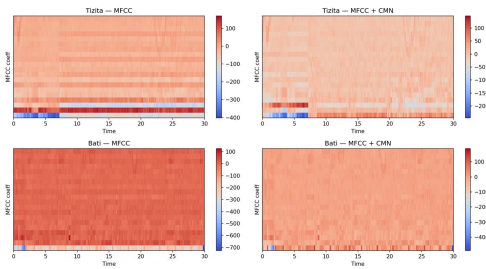


Fig. 1. MFCC Tizita Vs Bati

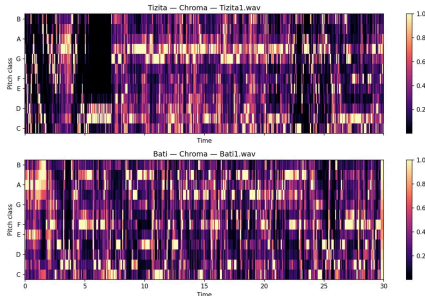


Fig. 2. Tizita and Bati Chroma (Replace this with real piano recording)

For each clip, the mean and standard deviation of all coefficients are computed over time, producing a fixed-length 64-dimensional feature vector. Stacking the feature vectors from all 306 recordings yields a 306×64 feature matrix. Although this representation discards temporal order, it is suitable for scale identification, where overall pitch and timbral characteristics are more informative than melodic or rhythmic structure.

C. Models

Two classifiers with different inductive biases are compared: an SVM with an RBF kernel (a geometric maximum-margin boundary) and a Random Forest of 200 trees (a rule-based ensemble with implicit feature selection). Both suit the small-sample, many-feature regime and are interpretable. As a diagnostic, k-means ($k = 2$) is also applied to the standardised features to test whether the scales separate without supervision.

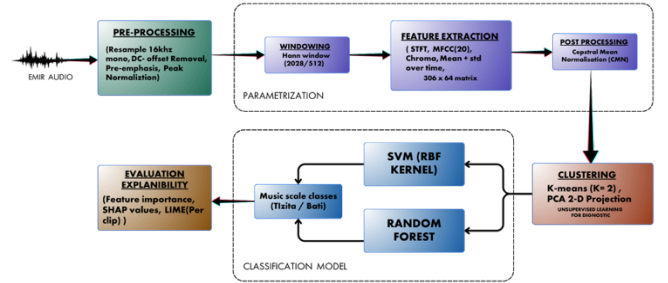


Fig. 3. Block diagram of the pipeline

III. EXPERIMENTS AND RESULTS

Phase I. Setup

Features were standardised (zero mean, unit variance) per fold. Classifiers were evaluated with stratified 5-fold cross-validation, with standardisation fitted inside each fold to avoid data leakage. Performance is reported as accuracy and macro-averaged F1 against a 52.9% majority-class baseline. Key parameters: MFCC $n=20$, chroma=12, $n_{fft}=2048$, $hop=512$; SVM $C=1.0$, $gamma='scale'$; Random Forest $n_{estimators}=200$.

Phase II. Unsupervised Analysis

k-means did not recover the scales: each cluster was a near-even mix of Tizita and Bati, and the same held for chroma-only and MFCC-only subsets. A 2-D PCA projection (Fig. 2) shows the classes thoroughly intermixed, indicating that scale is not the dominant source of variation; genre and recording conditions dominate. This does not preclude supervised separability, since a labelled model can target the boundary that clustering overlooks.

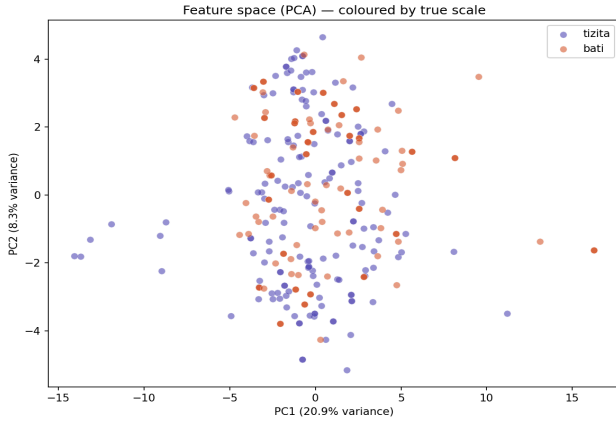


Fig. 4. PCA projection of the 64-dimensional feature space, coloured by true scale; the classes overlap rather than forming separate groups.

Supervised classifiers are required to assess learnability since apparent overlap in this view does not prohibit separability in the entire feature space, as PCA projection onto 2 components captures only ~29% of total variance and over 70% of the variation is in dimensions we can't see here.

Phase III. Classification

Because Random Forest and SVM (RBF kernel) represent essentially different classification algorithms, the comparison complements rather than duplicates. While the Random Forest integrates an ensemble of decision trees to represent complex feature interactions, the SVM learns a non-linear maximum-margin decision boundary. Because they can predict expected non-linear relationships across pentatonic scales, perform well in small-sample, high-dimensional contexts, and offer an interpretability level that supports the project's explainability target, both classifiers are well suited to this work.

Both classifiers exceeded the baseline substantially (Table I). The Random Forest reached about 86% accuracy and macro-F1, ahead of the SVM at about 80%. Accuracy and macro-F1 were nearly identical for both, indicating balanced performance across scales; the confusion matrices (Fig. 6) show strong diagonals with roughly symmetric errors.

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(venv) kaleabeseifu@Kals-MacBook-Pro ethiopian-scales % python confusion_matrices.py
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SVM				
	precision	recall	f1-score	support
bati	0.789	0.778	0.783	144
tizita	0.805	0.815	0.810	162
accuracy			0.797	306
macro avg	0.797	0.796	0.797	306
weighted avg	0.797	0.797	0.797	306

Random Forest				
	precision	recall	f1-score	support
bati	0.848	0.854	0.851	144
tizita	0.870	0.864	0.867	162
accuracy			0.859	306
macro avg	0.859	0.859	0.859	306
weighted avg	0.860	0.859	0.860	306

Fig. 5 Evaluation matrices for the SVM and Random Forest (cross-validated predictions).

Classifier	Accuracy	Macro-F1	Std. dev.
Majority baseline	0.529	—	—
SVM (RBF)	~0.80	~0.80	0.031
Random Forest	~0.86	~0.86	0.051

Table 1: Cross-validated classification performance

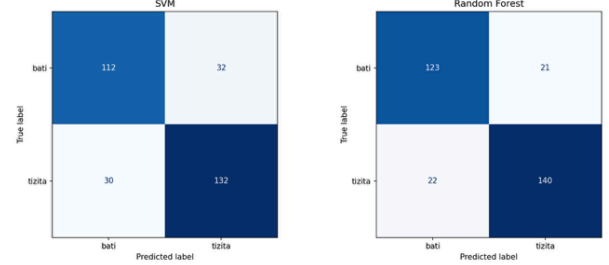


Fig. 6 Confusion matrices for the SVM and Random Forest (cross-validated predictions).

Per-fold accuracy (consistency) figure of the SVM and Random Forest under stratified 5-fold cross-validation shows: the dashed line marks the 52.9% majority-class baseline. The random forest is consistently at or above the SVM, with a higher mean (85.9% vs 79.7%) but greater fold-to-fold variation.

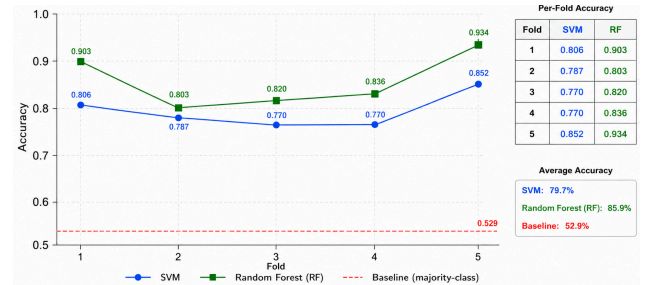


Fig. 7 Per-fold accuracy of the SVM and Random Forest

Comparing the results with existing work Retta *et al.* [1] report up to 95% with a deep CNN on the harder four-class task, and Ethiopian-genre studies report up to about 97% with LSTM models on larger datasets [3]. The present results are not directly comparable, being a two-class study on the most distinguishable pair using lightweight models; nonetheless, achieving about 86% with an interpretable Random Forest is a reasonable trade of accuracy for transparency and low cost.

Phase IV. Explainability

Three methods were applied to the Random Forest. Feature importance (Fig. 8) ranks the variability of the lowest MFCC (energy dynamics) as the strongest single predictor, with chroma and MFCC features interleaved thereafter. SHAP shows the direction of effects — high energy variability pushes predictions toward Bati, while several chroma features systematically distinguish the scales, evidencing genuine reliance on pitch-class content. LIME confirms the same pattern for individual clips. Together they indicate the model uses both timbral and tonal information; the prominence of an energy-related feature is noted as a possible link to recording or performance style, to be interpreted with caution.

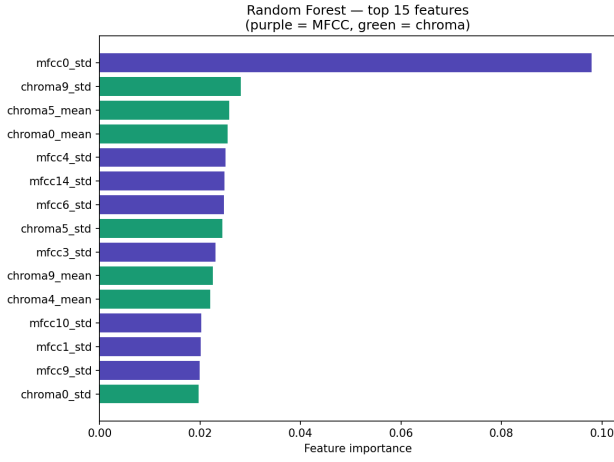


Fig. 8. Random Forest feature importance (top features; MFCC vs chroma indicated).

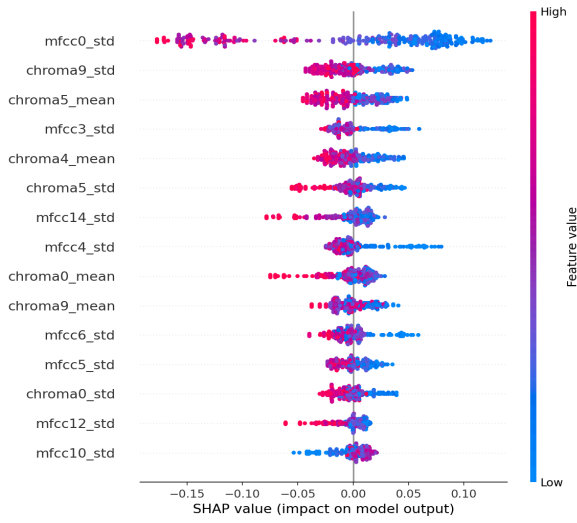


Fig. 9. SHAP analysis how features drive predictions.

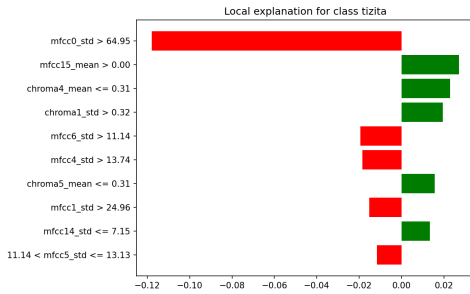


Fig.10. LIME explanation of an individual classification instance.

IV. LIMITATIONS AND FUTURE WORK

Several limitations bound the scope of this study and indicate directions for future work. First, only two of the four Kiñits are considered, and Tizita and Bati are among the most distinguishable [1]; the harder scales, notably Ambassel, are untested, so the reported accuracy is not directly comparable to four-class benchmarks. Extending to all four scales is the natural next step.

Second, summarising each clip by the mean and standard deviation of its features discards temporal order. Sequential and melodic cues that unfold over time are therefore lost; per-frame features with sequence models such as LSTMs, or shorter segmented windows (e.g. the 3

s segments used in [1], would retain this information and also increase the number of training samples.

Third, the genre/recording confound is observed but not resolved. The dominance of an energy-related feature (the variability of the lowest MFCC) suggests the model may exploit performance or recording style alongside genuine scale content; explicitly disentangling the two for example by testing cross-genre generalisation remains open. Relatedly, all data come from a single, modestly sized corpus (306 clips), and no evaluation on independent external recordings was performed, so generalisation beyond EMIR is unverified.

Fourth, standard 12-bin chroma assumes 12-tone equal temperament, whereas Ethiopian tuning is not strictly equal-tempered; constant-Q or microtonal-aware pitch features may represent the tonal content more faithfully. Finally, classifiers were used with default hyperparameters and STFT-based chroma; hyperparameter tuning and cleaner (CQT-based) chroma could improve performance, and bringing the present explainability analysis to deep models is a promising longer-term direction.

V. CONCLUSION

On EMIR, two Kiñits that are inseparable by unsupervised clustering can be classified by supervised models, with a Random Forest reaching about 86% accuracy showing scale information is present but subtle and entangled with genre and recording conditions comparing it to that of SVM. Beyond accuracy, the explainability analysis provides an interpretable account of the features driving classification, which prior deep-learning work on this dataset did not. Future work includes extending to all four Kiñits, segmenting clips into shorter windows for more data and direct comparability with [1], and applying interpretability methods to deep models.

VI. Code Availability

The complete source code, datasets preprocessing scripts, feature extraction modules, machine learning models, and explainability analysis are available at:

<https://github.com/seifukaleab/ethiopian-scales>

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